

Presented by Zhepei Wang

Adversarial Attacks on Automatic Speech Recognition

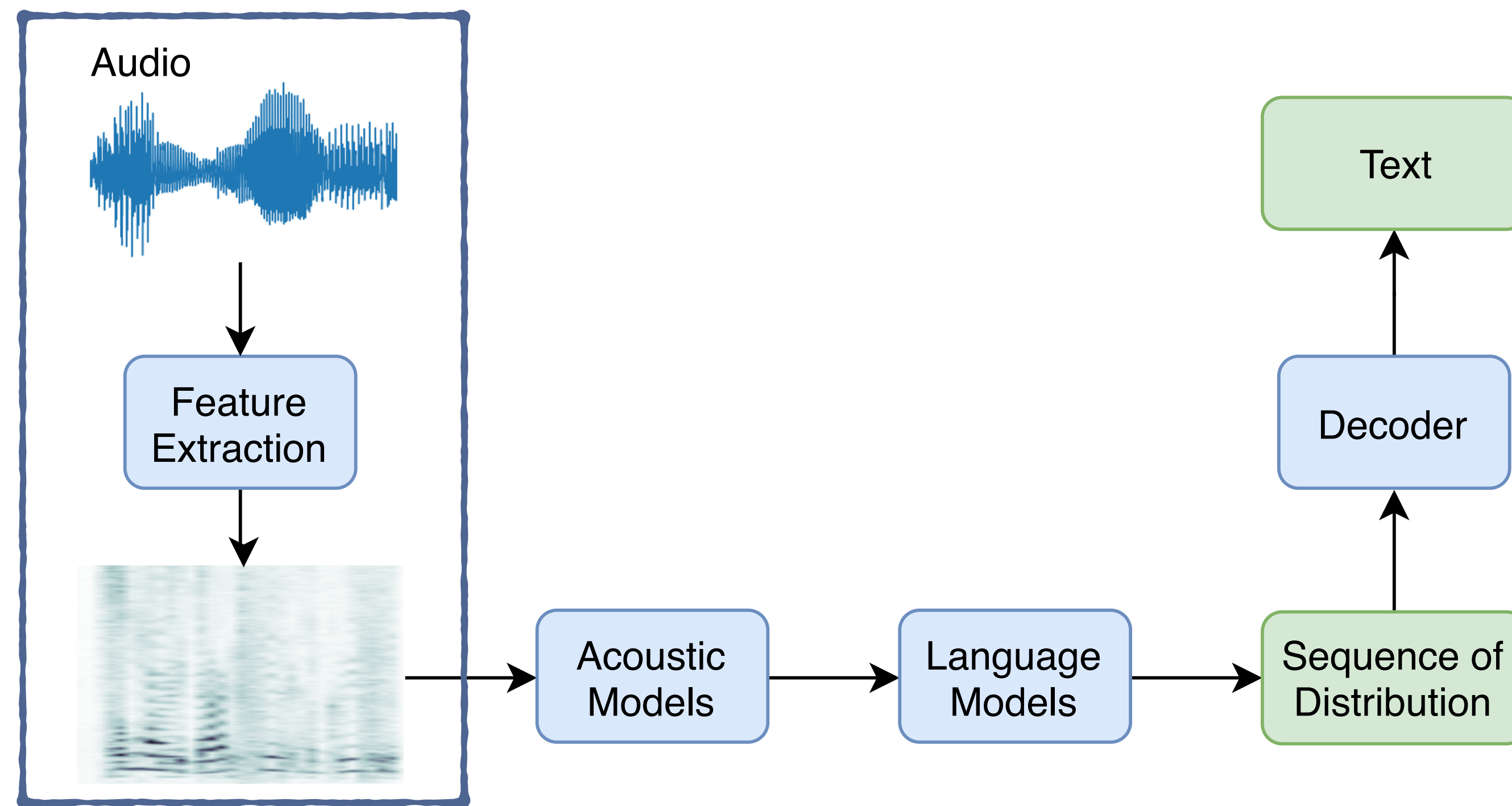
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Overview

- Background
 - Automatic Speech Recognition (ASR) framework
 - Adversarial attacks on audio
- End-to-end white-box targeted attack
 - “Audio Adversarial Examples: Targeted Attacks on Speech-to-Text”, Carlini et. al
- Attacks embedded in songs and with noise
 - “CommanderSong: A Systematic Approach for Practical Adversarial Voice Recognition”, Yuan et. al

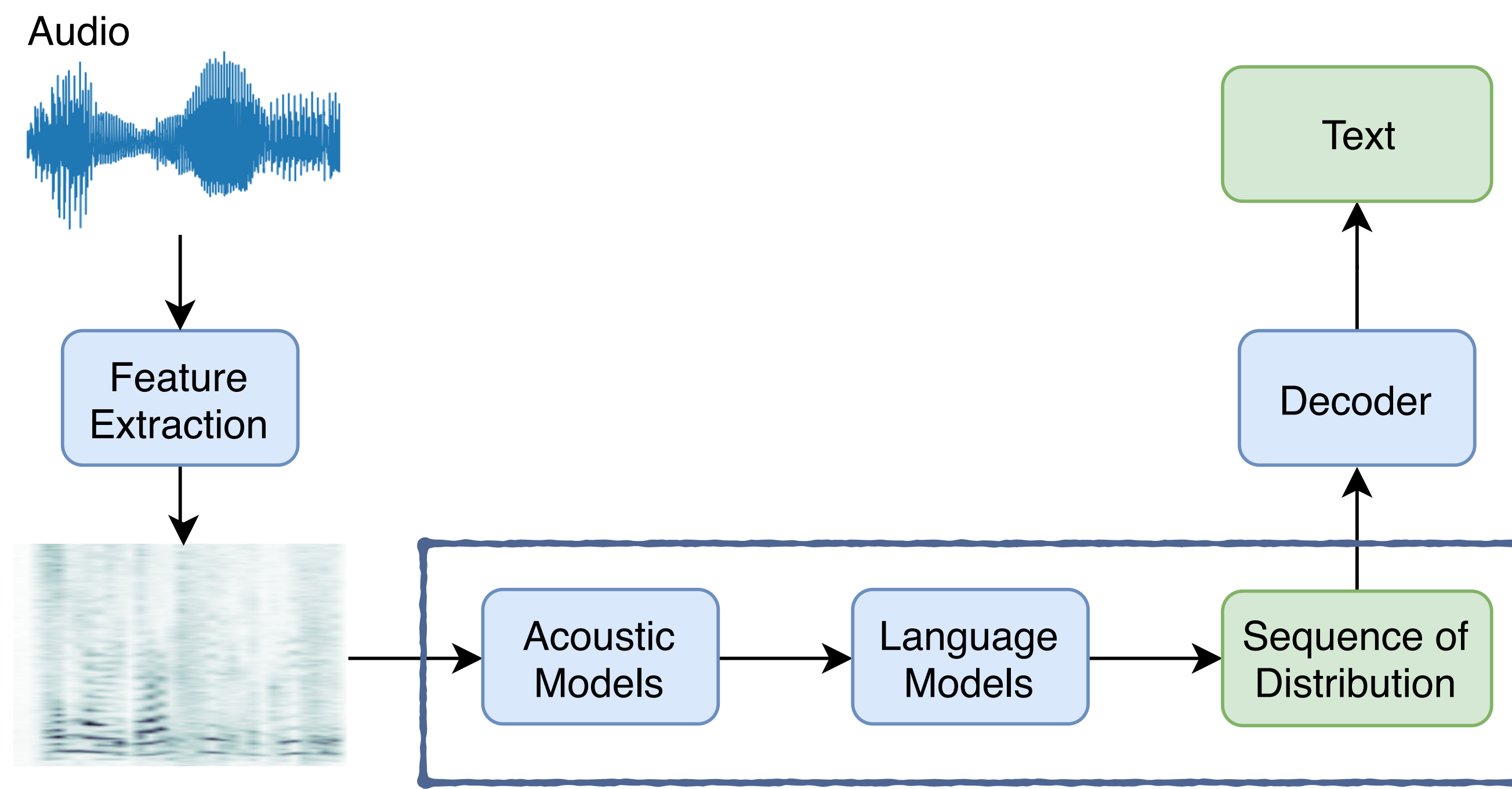
Automatic Speech Recognition System

- Input: time-domain 1-d vector $\mathbf{x} \in \mathbb{R}^T$
- Features: time-frequency domain Mel-Frequency Cepstral Coefficients (MFCC)



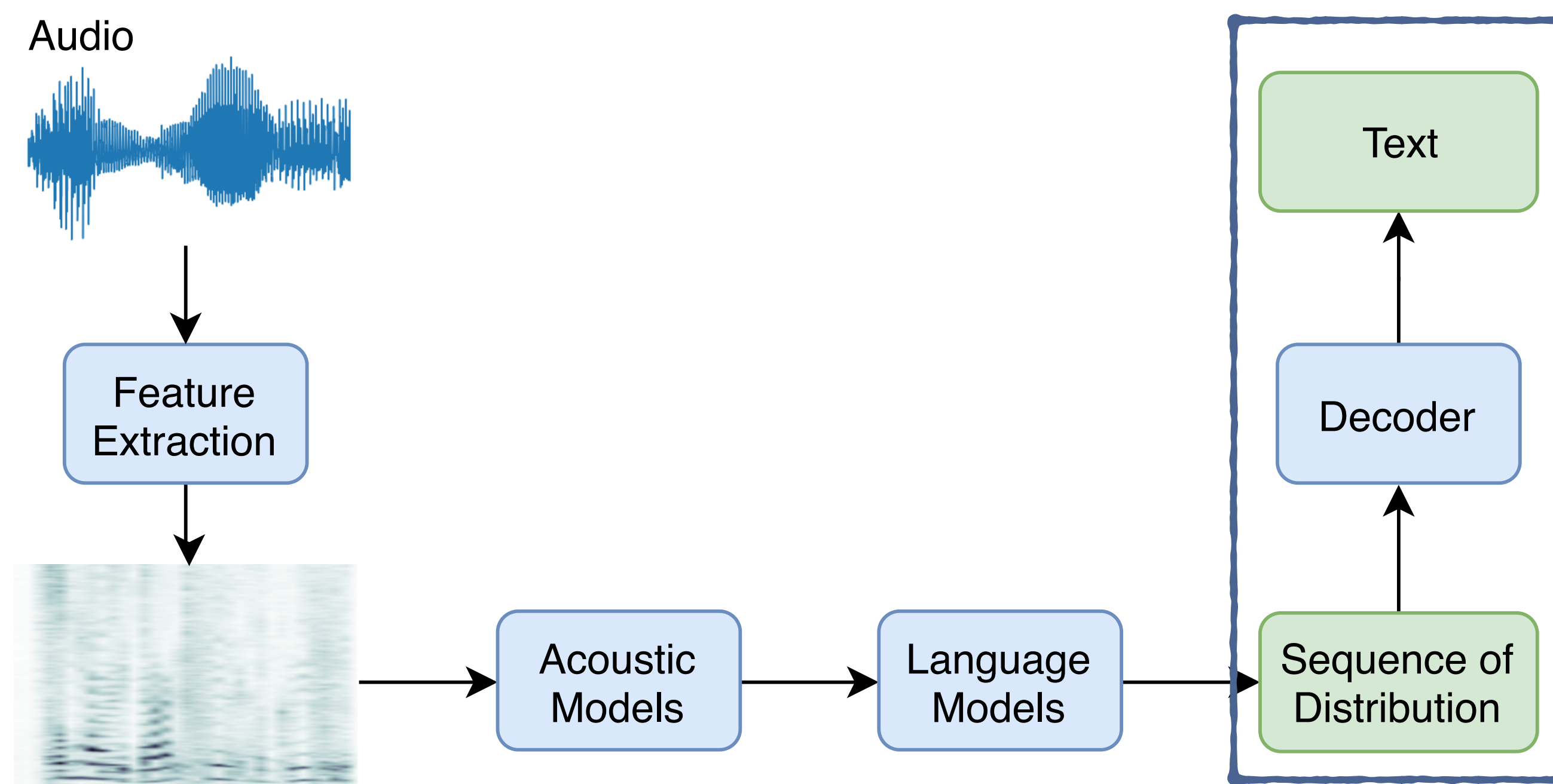
Automatic Speech Recognition System

- Acoustic/Language Models: GMM-HMM/RNN



Automatic Speech Recognition System

- Acoustic/Language Models: GMM-HMM/RNN
- Decoder: Greedy/Beam-search



Adversarial Attacks on ASR

- $\mathbf{x}' = \mathbf{x} + \delta \in \mathbb{R}^T, f(\mathbf{x}') \neq f(\mathbf{x})$
- Perturbation has to be imperceptible
- Challenges
 - High dimensionality in time-domain
 - Nonlinearity in MFCC
 - Different decoding algorithms
 - Ability to deliver in complex physical environment

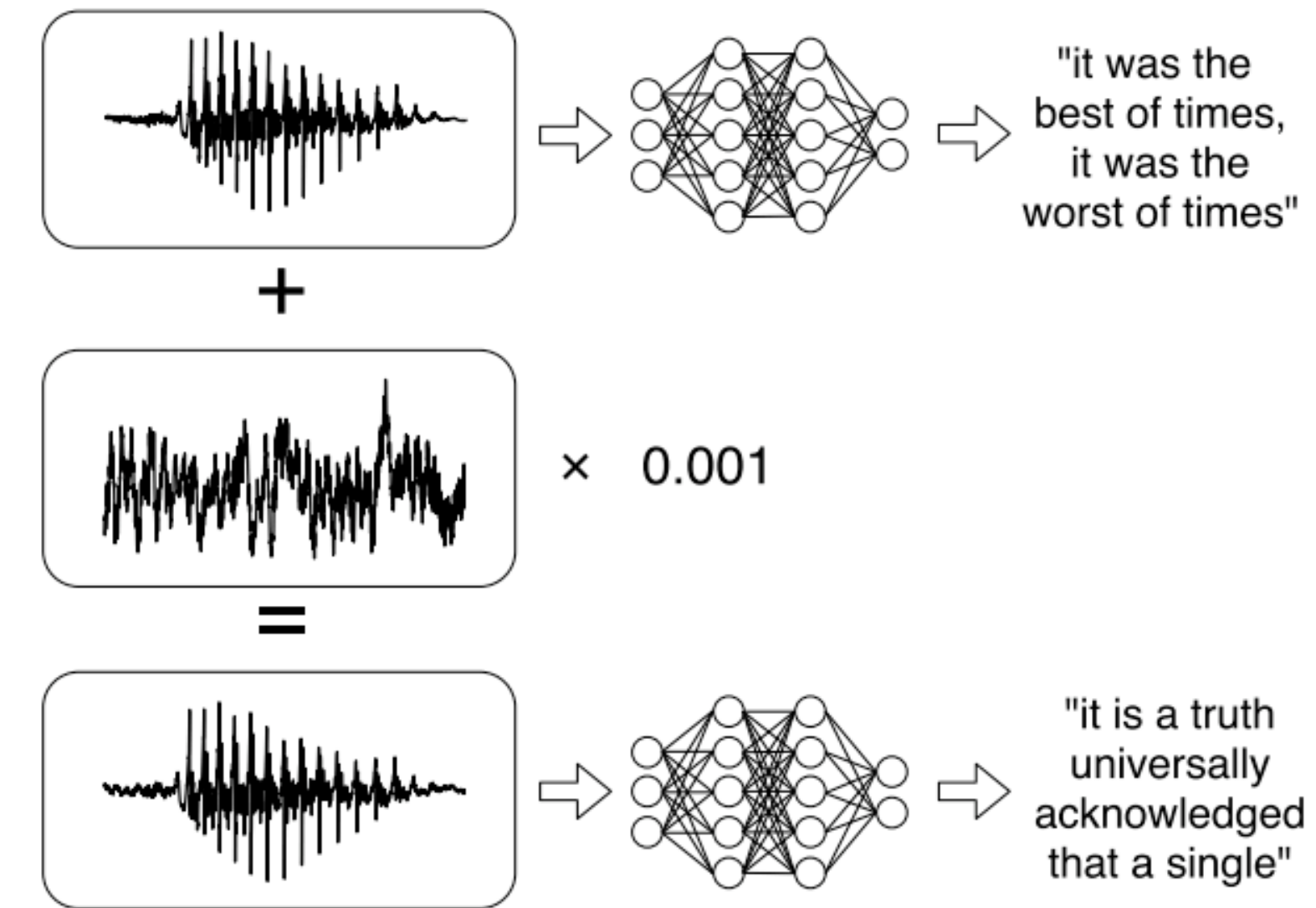


Figure adapted from Carlini et. al

Prior Studies

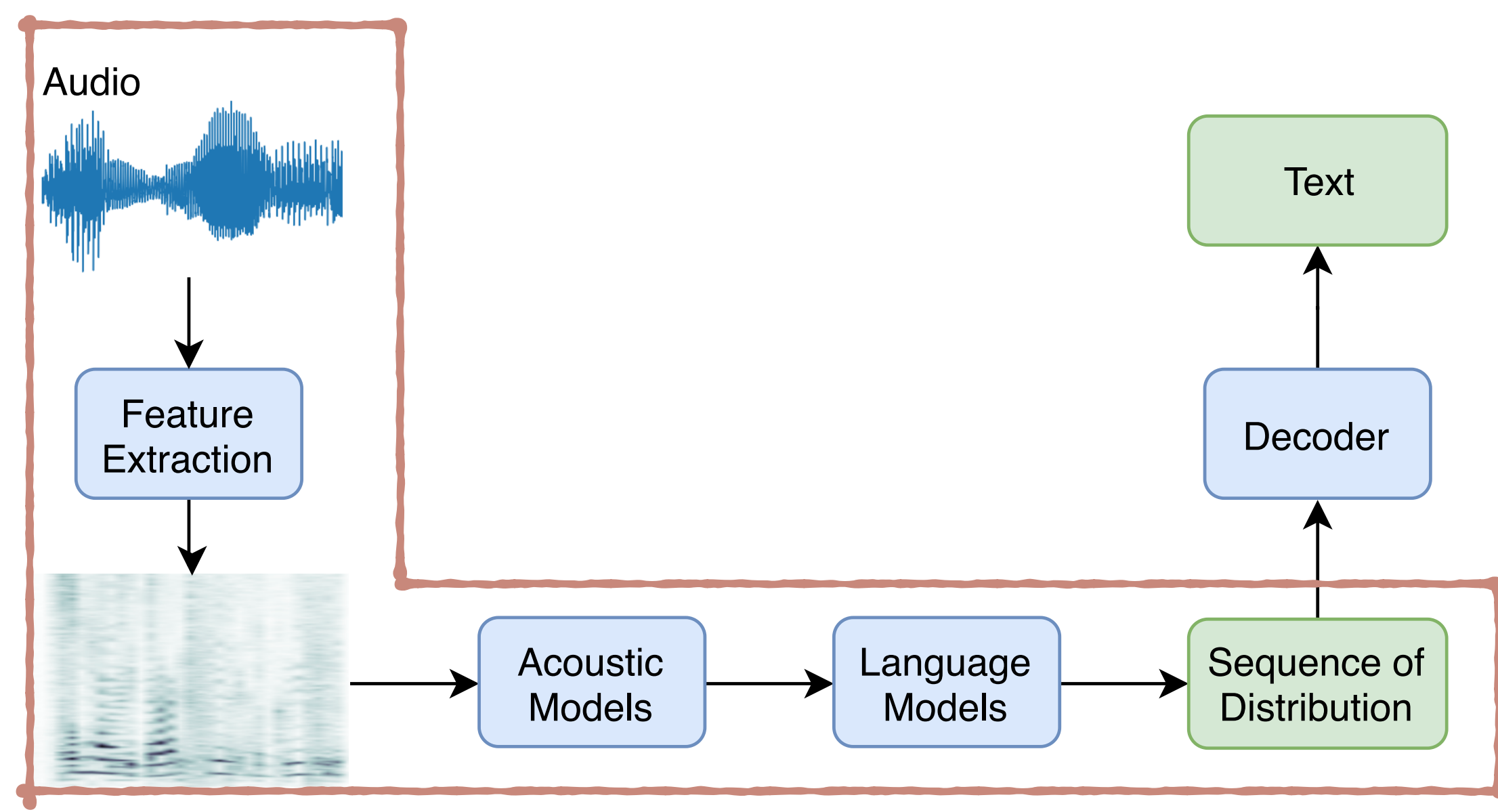
- Vulnerabilities in acoustic devices
 - Dolphinattack (Zhang et. al)
- GMM-based systems
 - Hidden Voice Commands (Carlini et. al)
- Targeted attacks on similar phrases
 - Houdini (Cisse et. al)

Targeted Attack on ASR

- White-box attack on DeepSpeech
- Targeted attack with arbitrary desired output
- Embedded in speech/non-speech
- Time-domain samples generated simultaneously

Recap: ASR Pipeline

- Output of DNNs is a sequence of character distribution
- Each input frame corresponds to an output frame
 - Length of output sequence may be different from the ground-truth sequence



Connectionist Temporal Classification (CTC)

- Idea: reducing longer sequence into a shorter one
- Repeating tokens for characters pronounced over one frames

Audio sequence

c c a a a t

Character sequence

c a t

Figure adapted from Hannun

Connectionist Temporal Classification (CTC)

- Idea: reducing longer sequence into a shorter one
- Repeating tokens for characters pronounced over one frames
- Blank token ϵ to help merge repeating tokens

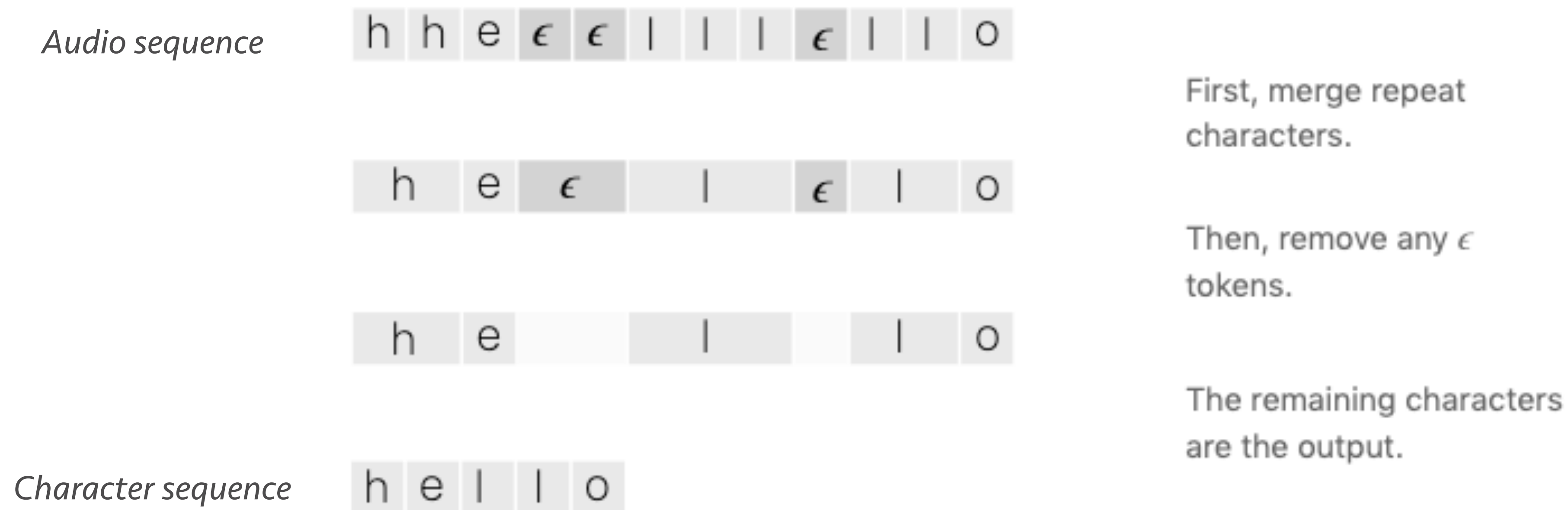


Figure adapted from Hannun

CTC Alignment

- Given
 - Ground-truth sequence $\mathbf{p}$
 - Model's output $\mathbf{y} = f(\mathbf{x}) \in \mathbb{R}^{V \times L}$
 - V is the vocab size and L is the number of frames
- A sequence π is alignment of $\mathbf{p}$ with respect to $\mathbf{y}$ if
 - π reduces to $\mathbf{p}$
 - $\text{len}(\pi) = \text{len}(\mathbf{y})$
- Alignment from π to $\mathbf{p}$ is many-to-one
 - Eg: $\mathbf{p} = ab, \mathbf{y} \in [0,1]^{3 \times 3}$
 - $\pi \in \{aab, abb, \epsilon ab, a\epsilon b, ab\epsilon\}$

CTC: NLL

$$\mathbb{P}(\mathbf{p} | \mathbf{y}) = \sum_{\pi \in \Pi(\mathbf{p}, \mathbf{y})} \mathbb{P}(\pi | \mathbf{y}) = \sum_{\pi \in \Pi(\mathbf{p}, \mathbf{y})} \prod_i y_i^{\pi_i}$$

$$\ell(f(\mathbf{x}), \mathbf{p}) = \text{CTC}(f(\mathbf{x}), \mathbf{p}) = -\log \mathbb{P}(\mathbf{p} | f(\mathbf{x}))$$

- Considers the probability under all possible alignments
- Implemented with dynamic-programming

Decoding

- Ideal

$$C^*(\mathbf{x}) = \operatorname{argmax}_{\mathbf{p}} \mathbb{P}(\mathbf{p} | f(\mathbf{x})) = \operatorname{argmax}_{\mathbf{p}} \sum_{\pi \in \Pi(\mathbf{p}, f(\mathbf{x}))} \mathbb{P}(\pi | f(\mathbf{x}))$$

- **Greedy**

$$C_{\text{greedy}}(\mathbf{x}) = \text{reduce}(\operatorname{argmax}_{\pi} \prod_{i=1}^L \mathbb{P}_t(\pi_i | f(\mathbf{x})))$$

- Beam-search

Distortion Metric

$$dB(\mathbf{x}) = \max_i 20 \log_{10}(x_i)$$

$$dB_{\mathbf{x}}(\delta) = dB(\delta) - dB(\mathbf{x})$$

Initial Formulation

$$\begin{aligned} &\text{minimize } \|\delta\|_2^2 + c \cdot \ell(\mathbf{x} + \delta, \mathbf{t}) \\ &\text{s.t. } dB_{\mathbf{x}}(\delta) \leq \tau \end{aligned}$$

- Using l2-norm for better convergence
- Starting with large τ and gradually decreasing its value
- Optimized with ADAM with a learning rate of 10 with maximum of 5,000 iterations

Initial Formulation: Issues

$$\begin{aligned} &\text{minimize } \|\delta\|_2^2 + c \cdot \ell(\mathbf{x} + \delta, \mathbf{t}) \\ &\text{s.t. } dB_{\mathbf{x}}(\delta) \leq \tau \end{aligned}$$

- CTC-loss penalizes labels that are already correct

$$\ell'(y, t) = \max(\max_{t' \neq t} y_{t'} - y_t, 0)$$

- c has to be large enough so that the most difficult character can be transcribed correctly
 - Different c_i 's for each frame π_i

Advanced Formulation

$$\begin{aligned} &\text{minimize } \|\delta\|_2^2 + \sum_i^L c_i \cdot \ell'(f(\mathbf{x} + \delta)_i, \pi_i) \\ &\text{s.t. } dB_{\mathbf{x}}(\delta) \leq \tau \end{aligned}$$

- First, solve with the initial formulation to obtain δ_0 and

$$\pi_0 = \operatorname{argmax}_{\pi} \prod_{i=1}^L \mathbb{P}_t(\pi_i | f(\mathbf{x}))$$

- Then, solve with the advanced formulation with $\pi = \pi_0$ and δ initialized to δ_0

Experimental Conditions

- Mozilla Common Voice dataset
 - First 100 test instances
 - For each instance, target 10 different transcriptions
- Evaluation
 - Success rate: success only if matches exactly the target phrase
 - Mean perturbation in dB

Experimental Results

- 100% success rate with mean distortion from -31dB to -38dB
 - Roughly equivalent to ambient noise in a quiet room
- Longer target phrases are more difficult
- Longer source phrases are easier to transform

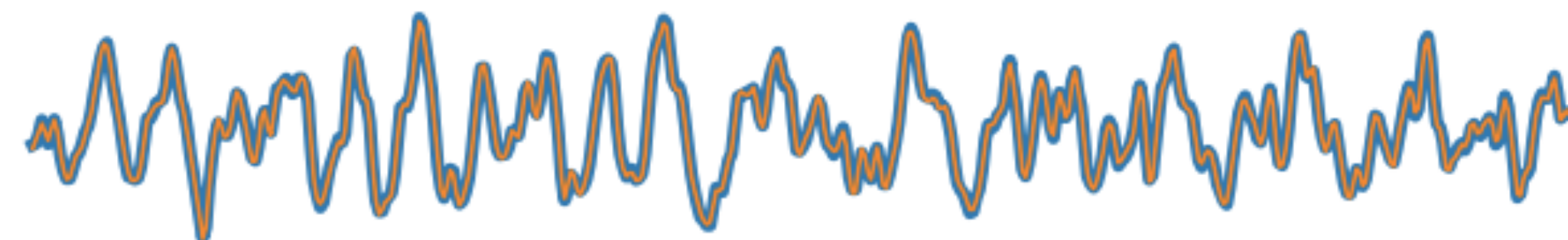


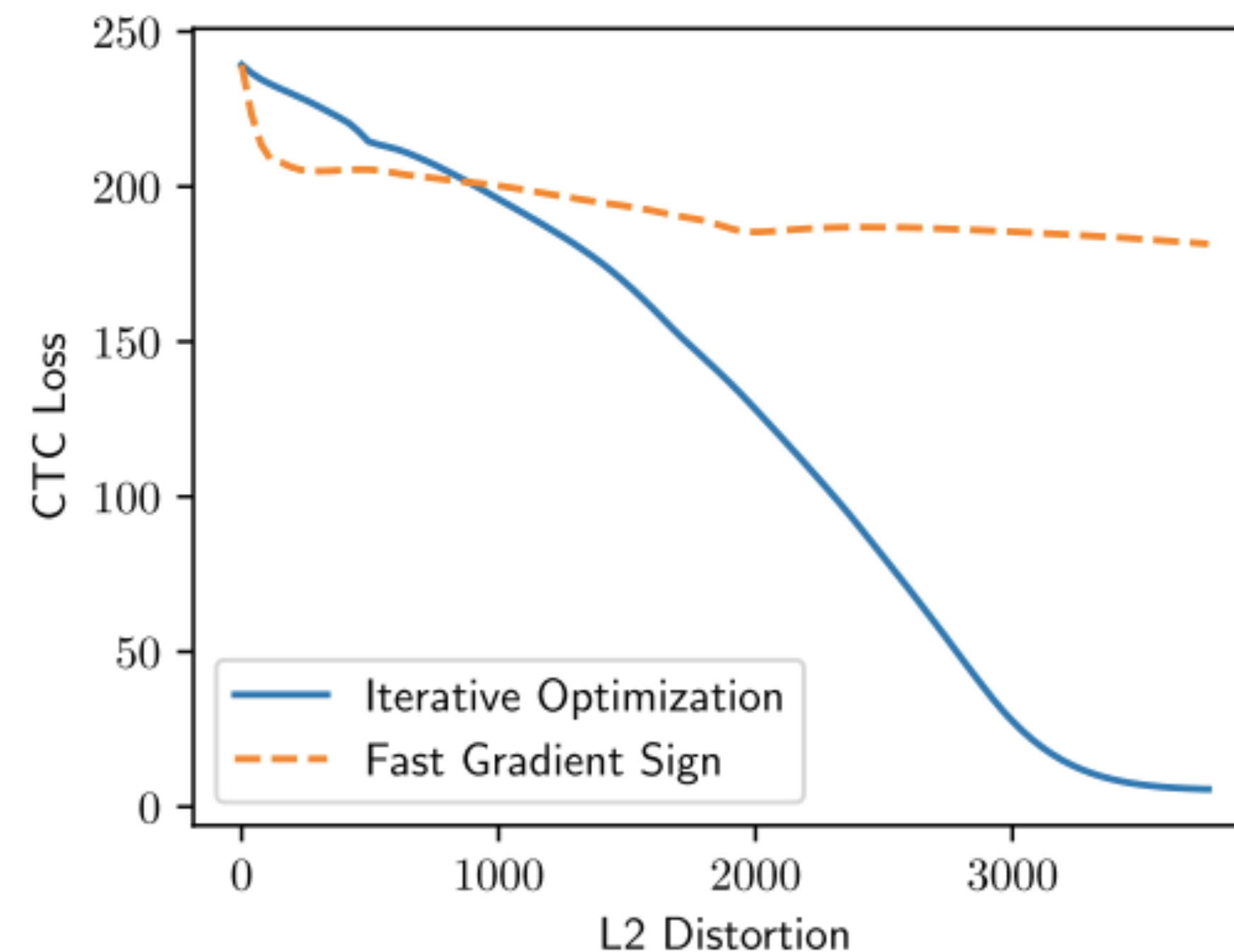
Figure adapted from Carlini et. al

Additional Experiments

- Shortest audio clip
 - $\text{len}(\pi) = \text{len}(\mathbf{p}) \implies \pi = \mathbf{p}$
 - No decoder involved
 - Still effective with mean distortion of -18dB
- Non-speech audio
 - Mean distortion of -20dB
- Targeting silence
 - Mean distortion of -45dB
 - Partially explains why longer source sequences are easier to transform
 - Silence frames not required and obtain the subsequence matching the target
 - For shorter source sequences, need to synthesize new frames to match the output

Properties of Generated Examples

- Comparison with FSGM (targeted)



- Nonlinearity in MFCCs and LSTMs make it challenging for FSGM
- Local linearity of NNs is not sufficient to generate targeted examples

Properties of Generated Examples

- Pointwise noise
 - Pointwise random noise will cause $\mathbf{x}'$ to lose its adversarial label
 - Expectation over Transforms may get around this problem with 10dB larger distortion
- MP3 Compression
 - Adversarial examples with approximately 15dB larger distortion

Takeaways

- Contributions

- End-to-end attack with arbitrary target sequences
- Alternative formulation to NLL-based loss
- Efficiency with non-speech audio and targeting silence

- Concerns

- Robustness under noise and ability to be transmitted under real-world conditions
- Studies of transferability

CommanderSong

- White-box attack on Kaldi
- “Hide” adversarial samples
 - Embed perturbations in a song
- Transmit in complicated physical environment
 - Noise modeling
- Impact a large amount of victims
 - Playing over video and radio

Attack Formulation

- Using Text-to-speech (TTS) tools to obtain command audio
- *pdf-id* sequence matching

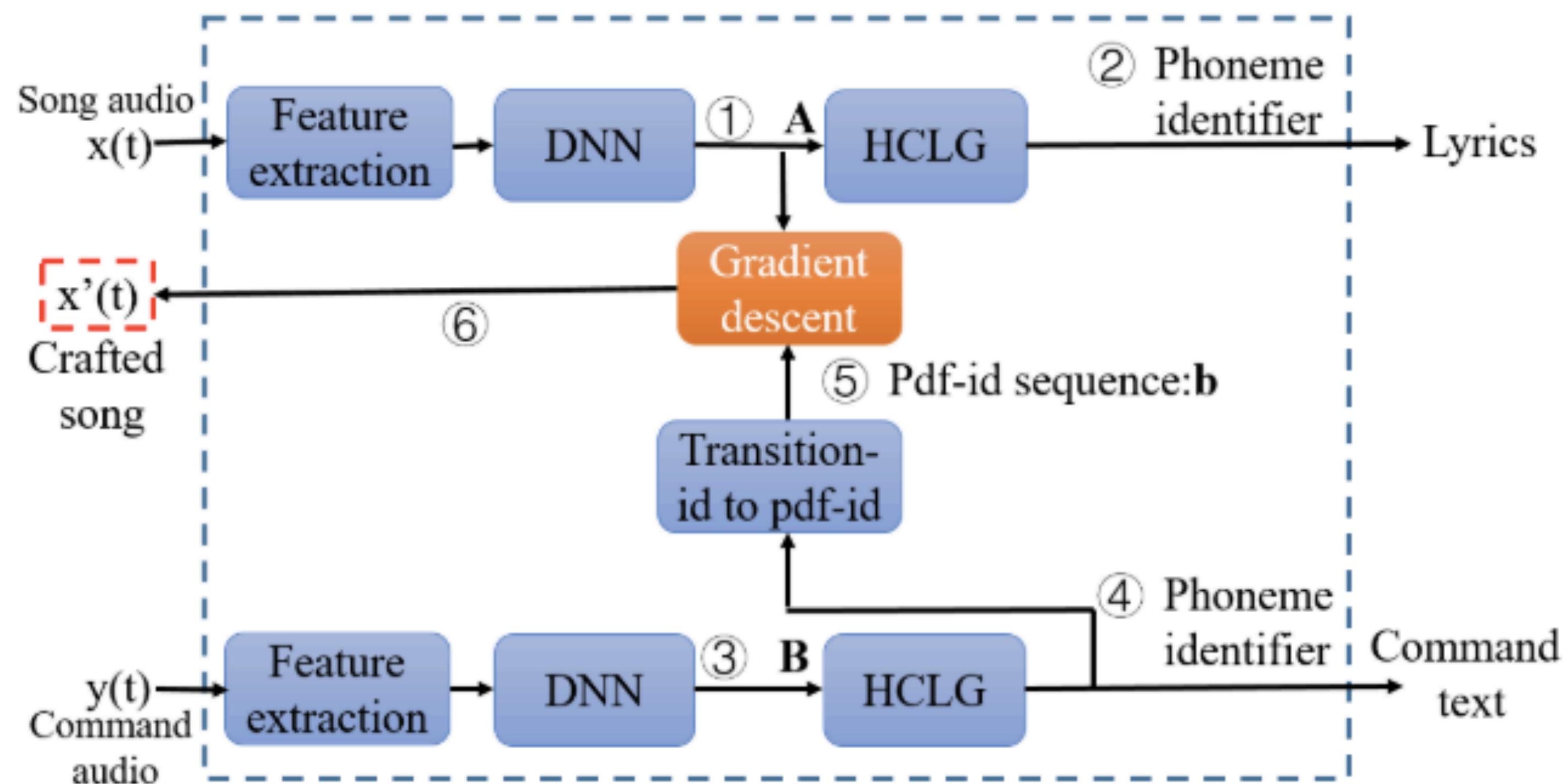


Figure adapted from Yuan et. al

pdf-id Sequence Matching

- pdf-id uniquely determines a phoneme with its transition state
- Let $\mathbf{A} = f(\mathbf{x}) \in [0,1]^{K \times N}$ be the DNN output for the song $\mathbf{x}$ with N frames and K pdf-ids
- $g(\mathbf{x})_i = \operatorname{argmax}_j A_{ij}$
- Let $\mathbf{b} = (b_1, \dots, b_N)$ be the highest probability pdf-id sequence for the command audio $\mathbf{y}$

Wav-To-API (WTA) Attack

$$\text{minimize}_{\delta} \|g(\mathbf{x} + \delta) - \mathbf{b}\|_1$$

$$\text{s.t. } |\delta| \leq \tau$$

- Aim to make $g(\mathbf{x} + \delta)$ close to $\mathbf{b}$ with minimal number of different phonemes

Wav-Air-API (WAA) Attack

$$\text{minimize}_{\mu} \|f(\mathbf{x} + \mu + \mathbf{n}) - f(\mathbf{y})\|_1$$

$$\text{s.t. } |\mu| \leq \tau$$

- Adversarial example is $\mathbf{x}' = \mathbf{x} + \mu$
- $n(t) \sim U(-N, N)$
- Not considering background noise since even the command $\mathbf{y}$ cannot be recognized by the system
- Major impacts come from the distortion of the receiver

Experiments: WTA

- 26 songs from Internet with different categories
- 12 sentences as commands
- Signal-to-noise ratio (SNR): $SNR = 10 \log_{10}(P_x(t)/P_\delta(t))$

Command	Success rate (%)	SNR (dB)	Efficiency (frames/hours)
Okay google restart phone now.	100	18.6	229/1.3
Okay google flashlight on.	100	14.7	219/1.3
Okay google read mail.	100	15.5	217/1.5
Okay google clear notification.	100	14	260/1.2
Okay google good night.	100	15.6	193/1.3
Okay google airplane mode on.	100	16.9	219/1.1
Okay google turn on wireless hot spot.	100	14.7	280/1.6
Okay google read last sms from boss.	100	15.1	323/1.4
Echo open the front door.	100	17.2	193/1.0
Echo turn off the light.	100	17.3	347/1.5
Okay google call one one zero one one nine one two zero.	100	14.8	387/1.7
Echo ask capital one to make a credit card payment.	100	15.8	379/1.9

Table adapted from Yuan et. al

Experiments: WAA

- Songs played with different speakers in a meeting room
- Audio received by an iPhone
- Testing with 2 of the 12 commands from WTA
- SNR significantly lower than WTA

Command	Speaker	Success rate (%)	SNR (dB)	Efficiency (frames/hours)
Echo ask capital one to make a credit card card payment.	JBL speaker	90	1.7	379/2.0
	ASUS Laptop	82	1.7	
	SENMATE Broadcast	72	1.7	
Okay google call one one zero one one nine one two zero.	JBL speaker	96	1.3	400/1.8
	ASUS Laptop	60	1.3	
	SENMATE Broadcast	70	1.3	

Table adapted from Yuan et. al

Transferability

- Kaldi -> iFLYTEK
 - Tested with three examples

Command	iFLYREC (%)	iFLYTEK Input (%)
Airplane mode on.	66	0
Open the door.	100	100
Good night.	100	100

Table adapted from Yuan et. al

- Kaldi -> DeepSpeech
 - DeepSpeech cannot correctly decode CommanderSong examples
- DeepSpeech -> Kaldi
 - 10 adversarial samples generated by CommanderSong (either WTA or WAA)
 - Modify with Carlini's algorithm until DeepSpeech can recognize
 - Modified samples successfully recognized by Kaldi with WTA attack

Experimental Results: Automated Spreading

- Online sharing
 - CommanderSong uploaded as video on YouTube
 - Bose Companion 2 speaker with iLFYTEK Input on LG V20
 - Command decoded successfully
- Radio broadcasting
 - CommanderSong broadcasted at FM 103.4 MHz
 - Radio setup at the corresponding frequency
 - iFLYTEK Input on several smartphones
 - Command always successfully recognized

Defense

- Audio turbulence

- Intuition: CommanderSong suffers from noise, but pure commands can still be recognized
- Compare results with and without applying noise
- Lower SNR indicates higher noise level
- WTA suffers from noise
- WAA is robust (since it's trained with random noises)

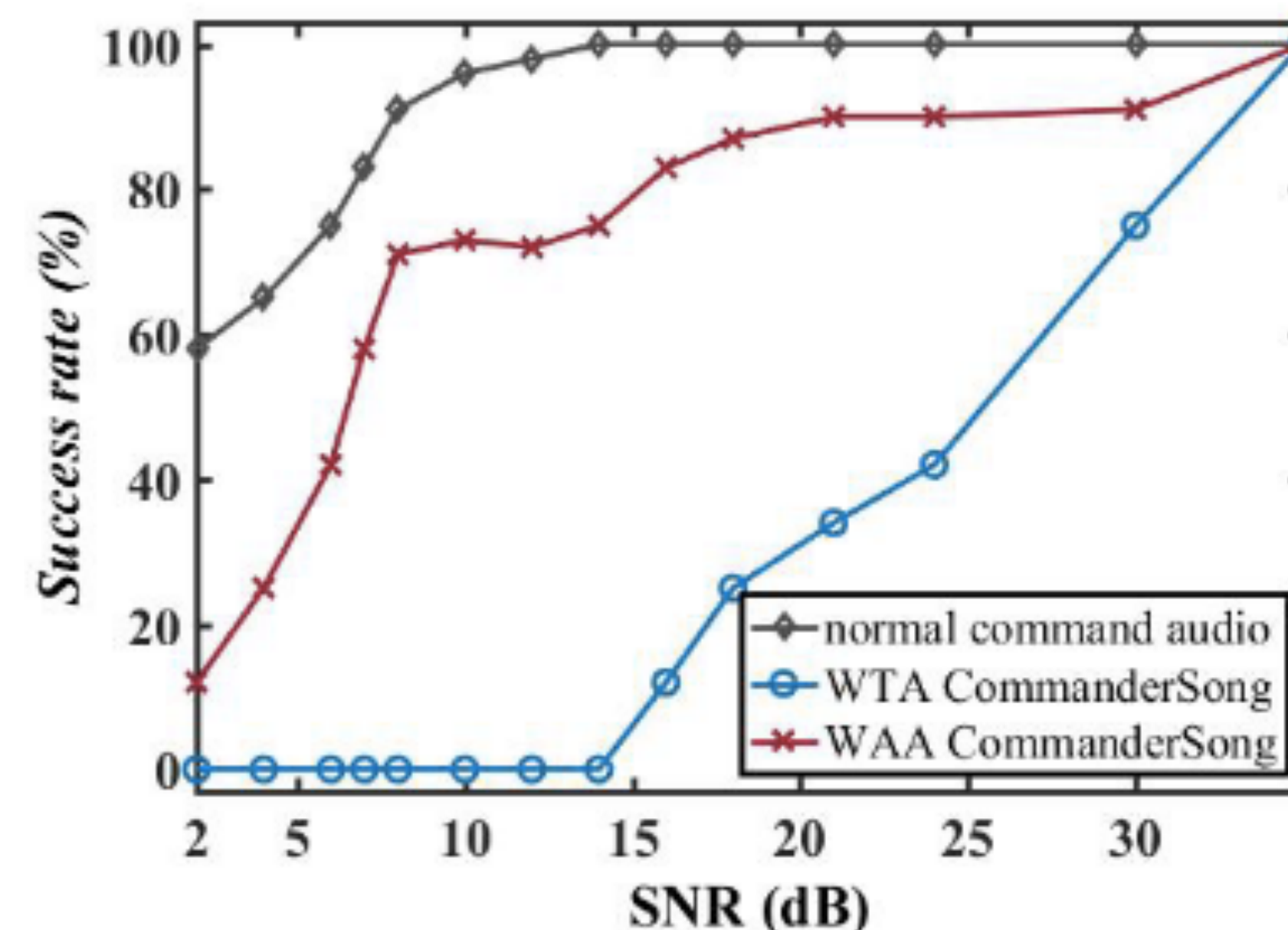


Figure adapted from Yuan et. al

Defense

- Audio squeezing
 - Downsample the input audio by a factor of M
 - Compare results with and without downsampling
 - Effective to defend against both WTA and WAA

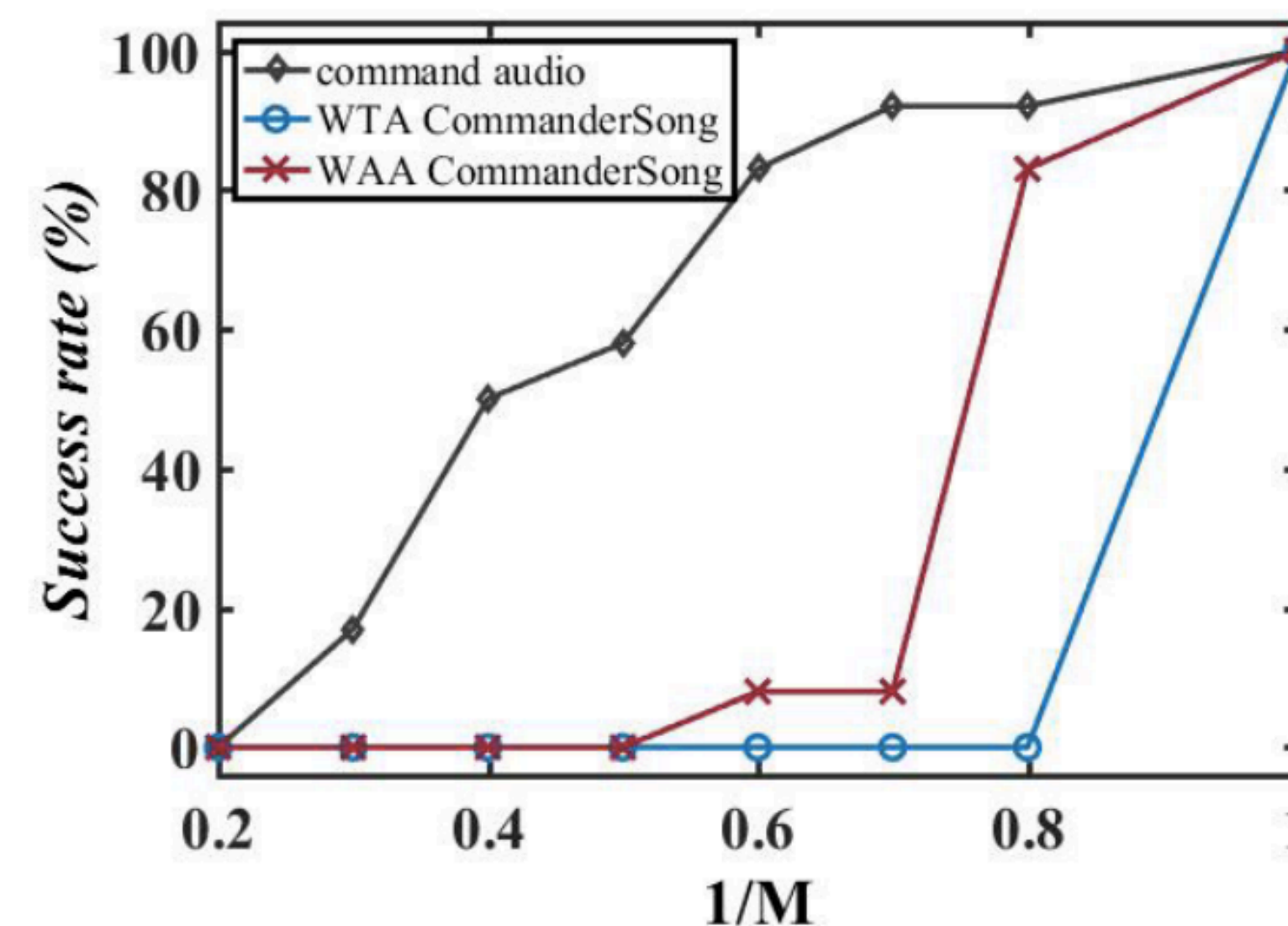


Figure adapted from Yuan et. al

Takeaways

- Contributions

- Embedding adversarial examples in songs
- Noise model to improve the robustness under random noise
- Ability to propagate via media
- Transferability between different ASR frameworks

- Concerns

- Oversimplified assumptions for noise
 - Alternative ASR models may be able to recognize pure voice commands with ambient noise
- Experimenting with different optimization strategies